

TRACE-rPPG: contactless heart rate from face video with Fourier series, the Fourier transform and convolution, and a measurement of how video compression interacts with skin tone

Signals and Systems course project | [Author names] | [Department, University] |
github.com/ahammadshawki8/TRACE-rPPG

Abstract. We build a remote photoplethysmography (rPPG) system in which every physiological number comes out of a convolution or a Fourier transform, with no trained model in the estimation path. The pipeline detrends by convolution with a moving average, band-limits with a hand-built windowed-sinc filter, locates the heart rate as the peak of a windowed FFT refined by parabolic interpolation, guards against harmonic lock-on using Fourier-series reasoning, and computes heart-rate variability with a second Fourier transform of the beat intervals. On top of green, CHROM and POS we build TRACE, a signal-quality-adaptive fusion with a pulse-blind artifact reference. We then ask a research question the literature has not answered: does video compression widen the known skin-tone accuracy gap? A simulated pilot of 36 volunteers across H.264, H.265, VP9 gives the following answer. For POS under H.264, the dark-minus-light error gap was 8.8 BPM on lossless video and 4.4 BPM at 100 kbps. The interaction coefficient was 0.40 BPM per doubling of bitrate (95% CI -0.57 to 1.37, $p = 0.420$; subject bootstrap -0.77 to 1.65). The interval includes zero: in this sample the curves do not fan apart detectably. Windows within 5 BPM: 79% lighter vs 56% darker on lossless video, 18% vs 15% at 100 kbps.

All results below come from a simulated pilot: real codecs, real detector and real pipeline applied to rendered faces whose skin-tone physics is modelled. They test the mechanism, not real people.

1. Introduction

Every heartbeat pushes blood into the capillaries of the face. Haemoglobin absorbs green light strongly, so the skin darkens very slightly with each beat: a change of well under one percent, invisible to the eye but within reach of an ordinary webcam [1]. Measuring it contactlessly matters wherever contact sensors fail: telehealth consultations, neonatal care, and driver monitoring. The difficulty is that the pulse is buried under lighting drift, motion and sensor noise that are all larger than it.

This project uses exactly the three tools of the course. **Convolution** removes drift and restricts the signal to plausible heart rates. The **Fourier transform** finds the rhythm, and a second Fourier transform of the beat intervals yields heart-rate variability. **Fourier series** explains why the spectrum has harmonics and why naive peak-picking can report twice the true rate.

2. Background and related work

Verkruyse et al. showed ambient light suffices [1]; Poh et al. used independent component analysis [2]; CHROM combines chrominance signals that cancel specular reflection under a standard skin colour [3]; POS projects onto the plane orthogonal to skin tone and weights its two axes adaptively [4]. Accuracy is known to fall on darker skin [5, 6] and under compression [7, 8]. The two are named as separately studied factors, and a search of 2,023 rPPG papers found none measuring their interaction. Deep models dominate recent work; we include two pretrained ones (PhysNet, FactorizePhys, via rPPG-Toolbox [9]) as baselines only.

3. Method

3.1 Where the course topics appear

Stage	Course topic	What it does
Face tracking	Fourier transform	Phase correlation: a translation multiplies the spectrum by a linear phase, so the inverse FFT of the normalised cross-power spectrum peaks at the shift.
Detrending	Convolution	Moving-average kernel (61 samples at 30 fps, first null 0.49 Hz); subtracting a low-pass leaves a high-pass.
Temporal normalisation	Convolution	Each channel divided by its running mean, turning skin colour into relative change.
Bandpass 0.7 to 4 Hz	Convolution	Hand-built windowed-sinc FIR (301 taps), zero phase; the convolution theorem is verified numerically.
Heart rate	Fourier transform	Hann-windowed FFT, zero-padded, parabolic interpolation between bins.
Harmonic check	Fourier series	A sharp pulse shape implies harmonics; before accepting a peak at f , test $f/2$.
Quality and fusion	Fourier transform (Parseval)	Fraction of in-band power at the peak and harmonic; artifact spectrum as a mask.
HRV, LF/HF	Fourier transform, again	Welch PSD of the RR series resampled to 4 Hz; LF 0.04 to 0.15 Hz, HF 0.15 to 0.40 Hz.
CHROM, POS	Linear algebra	Fixed projections from a skin reflectance model, not learned.

3.2 Verified on signals with known answers

Before any video, the mathematics was checked on synthetic pulses (12 of 12 checks pass). A 72 BPM pulse under lighting drift four times its size and noise of equal power was recovered as 72.07 BPM. Direct convolution and the FFT route agree to $4.3e-16$. Without zero padding the FFT computes circular convolution and corrupts the first $\text{len}(h)-1$ samples at 82 percent of scale. Parabolic interpolation cut the error on off-grid rates from 0.420 to 0.022 BPM. A pulse with a suppressed fundamental made naive peak-picking report 144 BPM; the sub-harmonic check recovered 71.99 with no false halvings from 86 to 140 BPM. The pulse was still recovered at -6 dB SNR.

3.3 From video to traces

Haar detection initialises the face box from the median of five detections; phase correlation then tracks it with a measured error of at most 0.13 px against the simulator's known motion. Re-running the detector every few frames made the box wobble by 1 to 2 px, which alone raised the green-channel error on dark simulated skin from 0.04 to 25 BPM, so the detector only corrects gross drift. The forehead and both cheeks are averaged with a per-frame adaptive chroma mask, so no fixed colour threshold favours one skin tone.

3.4 Extraction methods and TRACE

Green, ICA, CHROM and POS share one interface. After temporal normalisation a pure brightness change is the vector (1, 1, 1), which both CHROM and POS map to zero: measured output $2e-16$ and $3e-15$ against $1.8e-2$ for green. TRACE v1 weighted methods by spectral quality alone and lost to POS on held-out data, because a periodic motion artifact is also a concentrated peak (for green the quality score was inversely related to correctness). TRACE v2 adds a pulse-blind artifact reference whose spectrum masks each method's spectrum ($k = 4.0$) and penalises peaks shared with the artifact; weights are q^γ with $\gamma = 1.0$, and below quality 0.24 the reading is flagged low confidence. Parameters were tuned on one simulated cohort and frozen before testing on others.

3.5 Heart-rate variability

Beats are found on the POS pulse after a narrow band around the heart rate, then re-timed on the wideband waveform, because the narrow-band peak sits at the fundamental's phase rather than the systolic peak (that alone cost 41 ms RMS; re-timing brought it to 5.8 ms, under a fifth of a frame). Intervals are cleaned with the 20 percent rule and summarised as SDNN, RMSSD and pNN50. The RR series is resampled to a uniform 4 Hz grid before its Welch PSD, because the DFT requires uniform sampling. Checks: the integrated PSD equals the variance (Parseval), a two-tone RR series returns $(A_{lf}/A_{hf})^2$ and matches `scipy.signal.welch`, and captures under two minutes are refused for LF/HF because the LF band's period is 25 s. Choosing the waveform by spectral quality picked the motion-fooled green channel on half the subjects, so HRV uses POS. On clean simulated video the SDNN error was 0.7 ms and the RMSSD error 2.4 ms; with natural motion and screen light they rose to 36.7 and 61.6 ms, because a single missed or extra beat moves RMSSD by tens of milliseconds. Heart rate is robust; beat-to-beat variability from video is not, in this simulation.

4. Experimental setup

Simulator. A public-domain portrait is rendered with a two-layer reflectance model: melanin attenuates light more at short wavelengths, surface reflection is channel neutral, and the pulse modulates the dermal term along a standard blood-volume direction. Motion, lighting drift, chromatic light from a watched screen, sensor noise and a heartbeat with LF and HF rhythms complete it. The screen-light level was calibrated so uncompressed light-skin accuracy sits near published UBFC-rPPG figures.

Harness and grid. 36 simulated volunteers (18 per skin-tone group), 60 s each, encoded with H.264, H.265, VP9 at 100, 200, 400, 800, 1600 kbps with real-time settings, plus lossless RGB, YUV 4:4:4 and YUV 4:2:0 controls and a 4:4:4 H.264 ablation. The lossless control decodes bit-identically. Error is evaluated in 20 s windows with a 5 s hop against the reference PPG.

Statistics. Per-subject, per-condition MAE is modelled as $\text{err} \sim \log_2(\text{kbps}) \times \text{dark} + (1 \mid \text{subject})$. The interaction coefficient is the claim; it is reported with Wald and subject-bootstrap intervals and a log-error robustness check.

5. Results

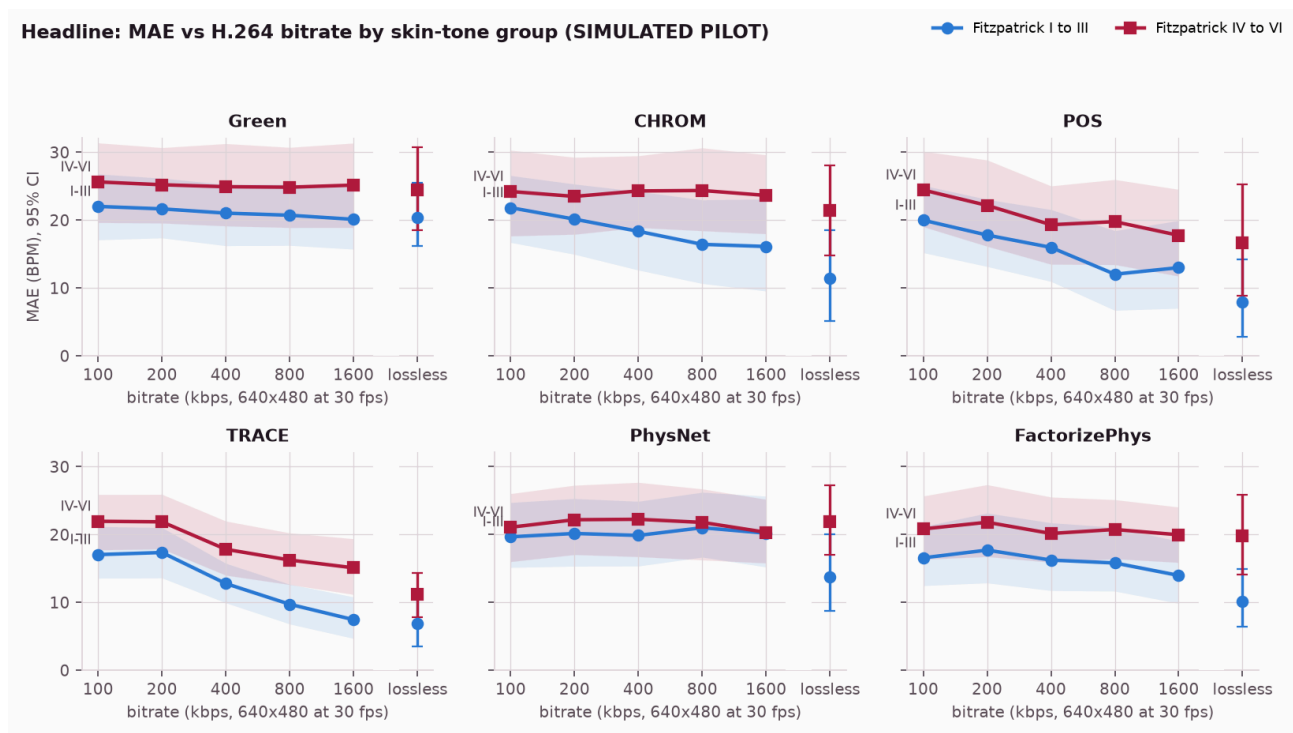


Figure 1. MAE against H.264 bitrate by skin-tone group; lossless at the right; 95 percent subject-bootstrap bands. Simulated pilot.

Headline. For POS under H.264, the dark-minus-light error gap was 8.8 BPM on lossless video and 4.4 BPM at 100 kbps. The interaction coefficient was 0.40 BPM per doubling of bitrate (95% CI -0.57 to 1.37, $p = 0.420$; subject bootstrap -0.77 to 1.65). The interval includes zero: in this sample the curves do not fan apart detectably. Windows within 5 BPM: 79% lighter vs 56% darker on lossless video, 18% vs 15% at 100 kbps.

CHROM. For CHROM under H.264, the dark-minus-light error gap was 10.0 BPM on lossless video and 2.4 BPM at 100 kbps. The interaction coefficient was 1.49 BPM per doubling of bitrate (95% CI 0.77 to 2.21, $p < 0.001$; subject bootstrap 0.43 to 2.35). The gap narrowed as bitrate fell, the opposite of the hypothesis. Windows within 5 BPM: 66% lighter vs 25% darker on lossless video, 14% vs 9% at 100 kbps. Darker skin was already near its failure plateau at every bitrate, and compression pulled lighter skin down to meet it: the narrowing reflects convergence at failure, not a benefit to darker skin.

MAE (BPM)	Lossless I-III	Lossless IV-VI	100 kbps I-III	100 kbps IV-VI
POS	7.9	16.6	20.0	24.4
TRACE	6.8	11.1	17.0	22.0
PhysNet	13.7	21.9	19.7	21.1
FactorizePhys	10.2	19.8	16.6	20.8

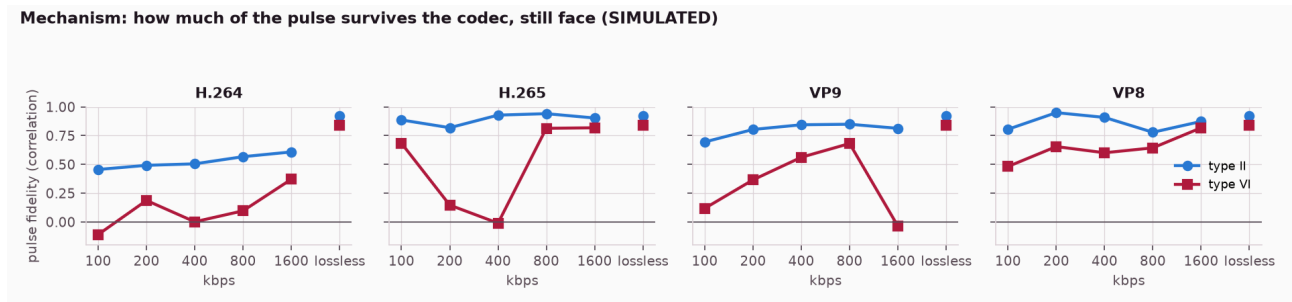


Figure 2. Pulse fidelity (correlation of the decoded green trace with the true pulse, still face, no screen light) after each codec and bitrate, lighter (II) and darker (VI) simulated skin. Lossless: 0.92 and 0.84; 4:2:0 subsampling alone: 0.93 and 0.09; H.264 at 1600 kbps: 0.61 and 0.37; at 100 kbps: 0.46 and -0.11. On a still face the codec also freezes regions that barely change, which removes the pulse outright.

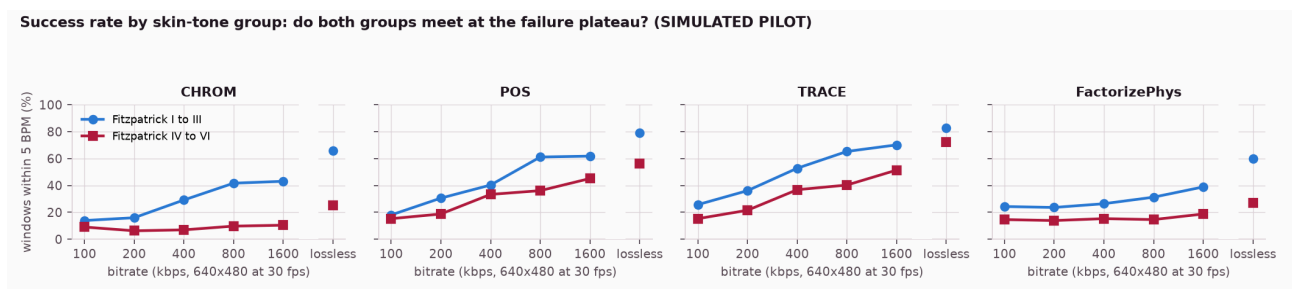


Figure 3. Share of windows within 5 BPM by group and bitrate. For POS: 79% lighter vs 56% darker on lossless video, 18% vs 15% at 100 kbps.

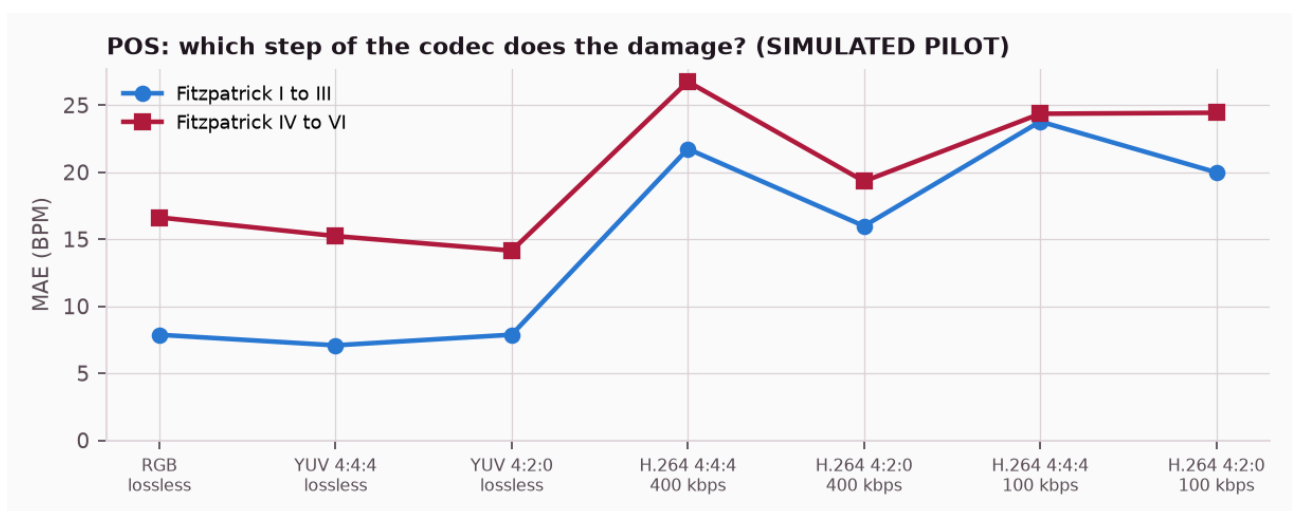


Figure 4. Which codec step does the damage, for POS: conversion, subsampling, quantisation, with a 4:4:4 control.

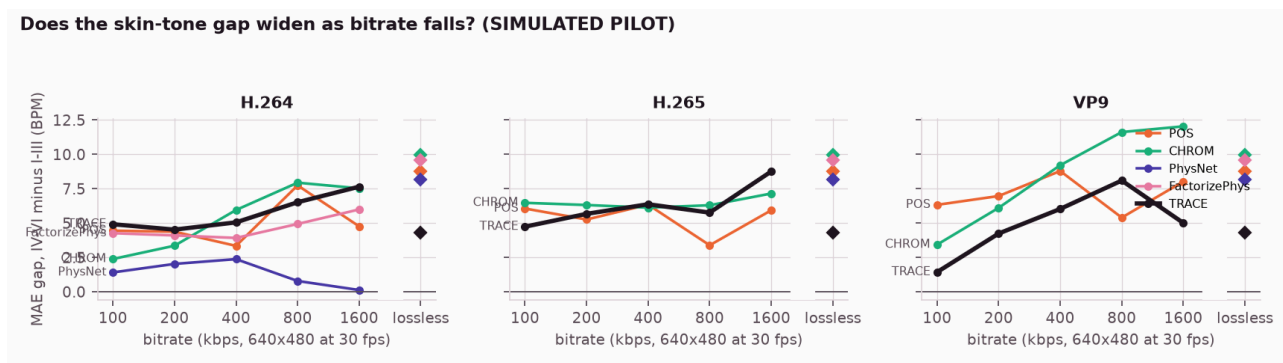


Figure 5. Dark-minus-light MAE gap against bitrate for each codec and method.

Mitigation. On the full grid the lowest error on lossless video came from TRACE (9.0 BPM) and at 400 kbps from TRACE (15.3). TRACE scored 9.0 and 15.3; POS 12.2 and 17.7; POS with the artifact mask alone 9.1 and 16.1; FactorizePhys 15.0 and 18.2. TRACE's lossless skin-tone gap was 4.3 BPM against 8.8 for POS. The interaction could not be estimated. On a held-out uncompressed cohort, POS scored 11.8 BPM, POS with the artifact mask alone 7.3, TRACE v1 19.8 and TRACE v2 9.7. Readings TRACE marked confident (93 percent of windows) had 8.7 BPM error against 22.8 for flagged ones.

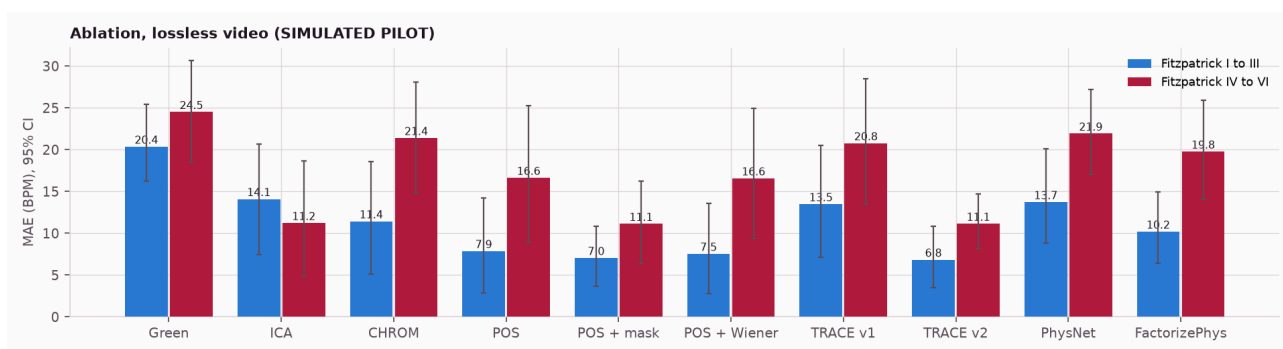


Figure 6. Ablation on lossless video, by skin-tone group.

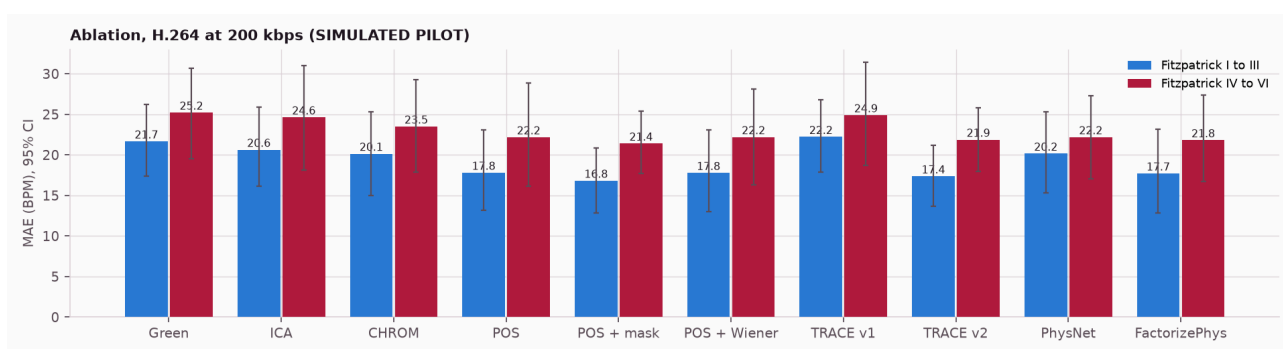


Figure 7. Ablation at 200 kbps H.264.

Bland-Altman by skin-tone group (SIMULATED PILOT)

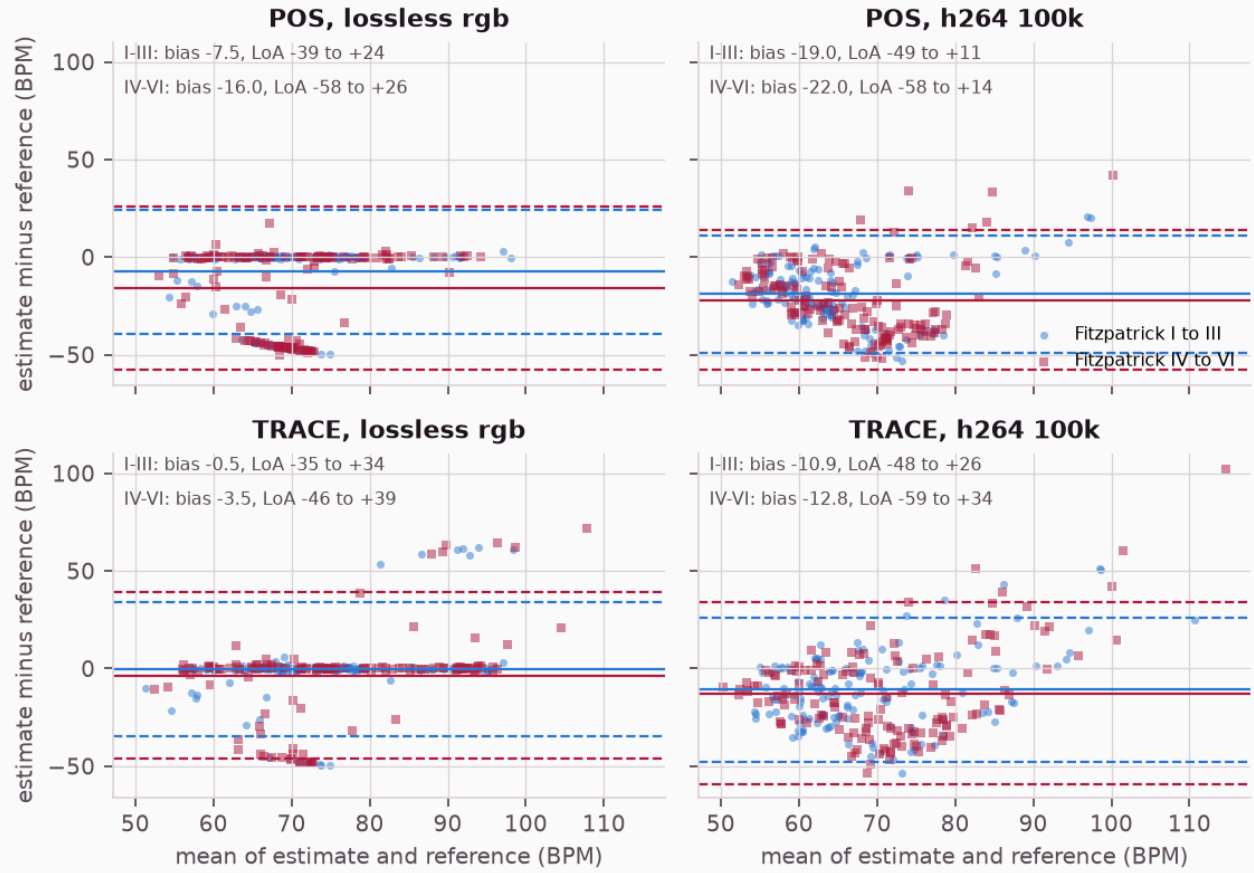


Figure 8. Bland-Altman plots by group, lossless and 100 kbps.

Learned baselines. PhysNet: For PhysNet under H.264, the dark-minus-light error gap was 8.2 BPM on lossless video and 1.4 BPM at 100 kbps. The interaction coefficient was -0.38 BPM per doubling of bitrate (95% CI -0.93 to 0.17, $p = 0.173$; subject bootstrap -1.01 to 0.25). The interval includes zero: in this sample the curves do not fan apart detectably. Windows within 5 BPM: 46% lighter vs 18% darker on lossless video, 13% vs 17% at 100 kbps. FactorizePhys: For FactorizePhys under H.264, the dark-minus-light error gap was 9.6 BPM on lossless video and 4.3 BPM at 100 kbps. The interaction coefficient was 0.43 BPM per doubling of bitrate (95% CI -0.15 to 1.01, $p = 0.145$; subject bootstrap -0.06 to 0.94). The interval includes zero: in this sample the curves do not fan apart detectably. Windows within 5 BPM: 60% lighter vs 27% darker on lossless video, 24% vs 15% at 100 kbps.

6. Discussion and limitations

Condition	Behaviour	Why
Low light	Fails	Needs reflected visible light.
Darker skin	Less accurate	Melanin lowers pulse amplitude in pixel levels; the subject of this study.
Heavy motion or talking	Green fails first; all degrade	Motion changes brightness and specular reflection in the heart-rate band.
Screen light	Sets an accuracy floor	Chromatic changes are not cancelled by CHROM or POS.
Low bitrate	Pulse attenuated	Chroma subsampling and quantisation of sub-level changes.
TRACE on light skin, low motion	Can lock on a harmonic	The artifact reference leaks pulse and the mask removes the fundamental.

The central limitation is that skin-tone physics is modelled: a simulated fan (or its absence) is evidence about the mechanism, not about people. The next step is recording volunteers across Fitzpatrick III to VI in Dhaka; the power analysis indicates more than 40 per group for 80 percent power at an interaction of 1.0 BPM per doubling of bitrate.

7. Ethics

The system never uploads video. Any recording of people requires informed consent, coded storage and deletion after the study (protocol in `deliverables/data_collection_protocol.md`). Heart-rate variability is a wellness indicator and never a diagnosis. Skin-tone and lighting limitations are reported rather than omitted.

8. Future work

Estimate the pulse direction per window from the data so the artifact reference cannot leak pulse; validate on real lossless recordings; extend to camera-only ballistocardiography, whose failure modes are nearly orthogonal to rPPG's.

References

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